Random Forest — Model Validation

Label = CDI stage · split train <= 2024 / test > 2024 (year-based) · test accuracy 77.6% (macro F1 0.607) · saved model, inference only

(a) Per-class metrics (test set)

Class	Precision	Recall	F1	Support
0 No Drought	0.950	0.776	0.854	98
1 Mild	0.703	0.821	0.757	78
2 Moderate	0.167	0.286	0.211	7
3 Severe	–	–	–	0
4 Extreme	–	–	–	0
Accuracy	**0.776**	**0.776**	**0.776**	**0**
Macro avg	**0.607**	**0.627**	**0.607**	**183**
Weighted avg	**0.815**	**0.776**	**0.788**	**183**

(b) Train / test summary

Rows with full features	1075 (130 months x 9 tehsils - cloud-gap rows)
Split	train <= 2024 / test > 2024 (year-based)
Train / test	892 / 183 (n_test = confusion-matrix total)
Class balance (train)	S0:513, S1:303, S2:55, S3:21
Model	RandomForestClassifier(n_estimators=100, max_depth=10, class_weight='balanced', random_state=42)
Features	SPI-1, SPI-3, SPI-6, SPI-12, SSMI, NDVI, EVI, PCI
Labels	CDI stage; SPI-3 stage (multi-output)
SPI-3 stage accuracy	90.7% (expected high: SPI-3 is itself a feature)

LEAKAGE CAVEAT: CDI is built from SPI-3, SSMI, VCI(NDVI), EVI and PCI - the very variables used as RF features - so a high accuracy is expected and is NOT independent skill.